

STAT 3113 - 104
Fall 2024
Exam 3

Name: _____

Instructor: Jarrad Botchway

Instructions:

1. Be sure to print your name in the space provided at the top of this page.
 2. Turn off your cell phone.
 3. There are **6 problems**. Point values are indicated (in parentheses) for each problem. The total number of points is 100.
 4. There are 8 pages in this set of Exam 3 (including this page). **Please check to make sure you have 8 pages with 6 problems.**
 5. For questions that require you to put final answers in boxes, be sure to do that.
 6. Please ask only your proctor if you have any questions during the exam.
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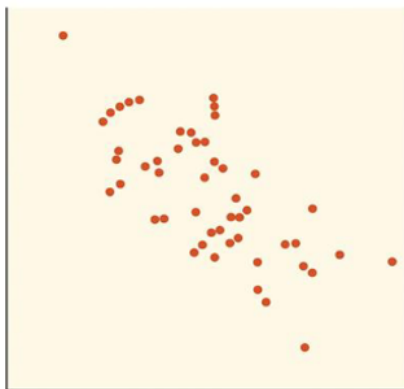
Question:	1	2	3	4	5	6	Total
Points:	12	10	6	17	30	25	100
Score:							

1. (12 points) *For the problems on this page, shade the correct answer.*
- (a) (2 points) I want to examine the relationship between the Class Attendance (number of times students attend class) and Test 2 Score for this semester. Which graphical display would show how Test 2 Score vary with Class Attendance?
- ☐ Boxplot
 - ☐ Density curve
 - ☐ Histogram
 - ☒ Scatterplot
- (b) (2 points) Suppose we are interested in the correlation r between Class Attendance (x) and Test 2 Score (y). Which statement about the correlation r is false?
- ☒ r is a number between -1 and 0
 - ☐ r describes linear relationships only
 - ☐ r is the same between Class Attendance and Test 2 Score as it is between Test 2 Score and Class Attendance
 - ☐ r does not have any units attached to it
- (c) (4 points) Data was collected on Class Attendance (x) and Test 2 score (y) and the following least squares regression line was fitted to the data. The resulting equation is $\hat{y} = 76.25 + 0.81x$. What is the residual for the point where $x = 12$ and $y = 100$?
- ☐ -57.25
 - ☐ -14.03
 - ☒ 14.03
 - ☐ 57.25
- (d) (2 points) In the simple linear regression equation $\hat{y} = 76.25 + 0.81x$, 0.81 is the _____.
- ☐ value of y when x is equal to 1
 - ☐ point where the line intersects the x -axis
 - ☒ change in y due to one-unit change in x
 - ☐ change in x due to one-unit change in y
- (e) (2 points) If we are interested in determining whether Class Attendance (x) and Test Score (y) are linearly related, it is necessary to _____.
- ☐ perform a t-test on the intercept β_0
 - ☒ perform a t-test on the slope β_1
 - ☐ perform a t-test on the pearson's correlation r
 - ☐ none of the above

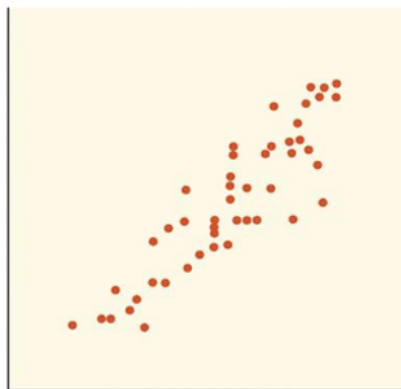
2. (10 points) **Shade the correct answer in the statements below.**

- (a) ☒ True ☐ False A balanced design is one where the number of replicates is same for each treatment.
- (b) ☐ True ☒ False ANOVA is a statistical measure adopted to analyze differences in group variances.
- (c) ☐ True ☒ False The results from a two-factor analysis of variance show a significant main effect for factor A and a significant main effect for factor B. Based on this information, there is significant interaction between A and B.
- (d) ☐ True ☒ False Every outlier is an influential point, but not every influential point is an outlier.
- (e) ☒ True ☐ False An example of a lurking variable could be the weather when studying the relationship between ice cream sales and shark attacks.

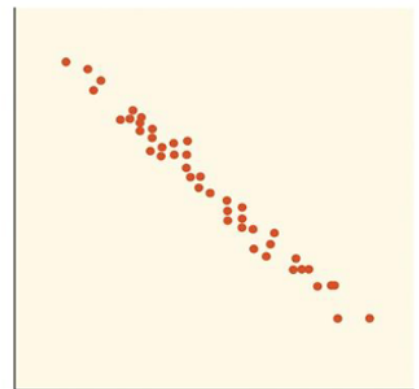
3. (6 points) **MATCHING:** Complete the table below by matching each of the following correlation (r) values to the appropriate scatterplot: $\{-0.99, -0.7, 0.9\}$



Plot A



Plot B



Plot C

Plot	A	B	C
Correlation r	-0.7	0.9	-0.99

4. (17 points) A three-way ANOVA is conducted to understand if there is an interaction between education, degree type and gender on salary. The levels of education are undergraduate, masters and PhD, the levels of degree type are business studies, psychology, biological sciences, engineering and law and the levels of gender are male and female. The effect test p-value output is given below. Using $\alpha = 0.05$, give appropriate conclusions.

Part I: (3 points) How many treatment combinations are there for this experiment?

$$3 \times 5 \times 2 = 30$$

Part II: Answer the questions below using I, II or III in the space provided.

I. Should be tested. Significantly impacts the mean salary

II. Should be tested. Does not significantly impact the mean salary

III. Should not be tested

Source	p-value
Degree type	0.2987
Education	0.0000
Gender	0.4476
Degree type * Education	0.0000
Degree type * Gender	0.0036
Education * Gender	0.0016
Degree type * Education * Gender	0.1056

(a) (2 points) The three-way interaction term Degree type * Education * Gender II

(b) (2 points) The two-way interaction term Degree type * Education I

(c) (2 points) The two-way interaction term Degree type * Gender I

(d) (2 points) The two-way interaction term Education * Gender I

(e) (2 points) The main effect of Degree type III

(f) (2 points) The main effect of Education III

(g) (2 points) The main effect of Gender III

5. (30 points) A study is conducted to investigate the relationship between average daily temperature and the total number of bike crossings over bridges in New York City. A simple linear regression model is fitted. Use the outputs on **page 6** to answer the following questions.

- (a) (6 points) Identify the following for this problem.

Response variable Total number of bike crossings

Explanatory variable Average daily temperature

- (b) (4 points) What is the least-squares regression equation for this problem? Report only 3 decimal places of the coefficients.

$$\widehat{\text{Total-Crossings}} = -5264.68 + 370.10 \text{ Avg-Temp}$$

- (c) (5 points) According to the fitted line, what will be the number of bike crossings on average, when the average daily temperature in New York City is 55F ?

☐ 5091 ☒ 15091 ☐ 17000 ☐ 19051

- (d) (5 points) The proportion of the variability in number of bike crossings over bridges in New York City accounted for by average daily temperature is

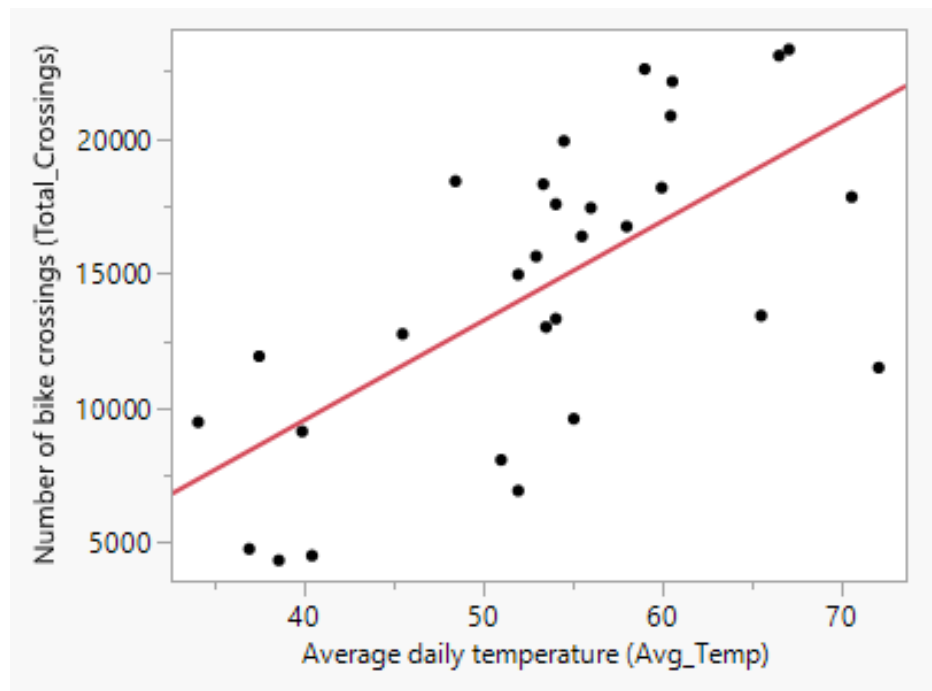
☐ 37.01% ☐ 42.92% ☒ 43.19% ☐ 52.65%

- (e) (5 points) If the average daily temperature in New York City goes up by 5 F, the line predicts the total number of bike crossings to _____. (when rounded to the nearest whole number)

☒ increase by 1851
☐ decrease by 1851
☐ increase by 370
☐ decrease by 370

- (f) (5 points) Should the intercept of the line be interpreted in this case?

☐ Yes, as the average number of bike crossing when average daily temperature in New York City is zero F
☐ Yes, as the average daily temperature in New York City when there are no bike crossings
☐ No, since number of bike crossing of 0 is out of the range observed
☒ No, a zero F average daily temperature in New York City is out of the range observed



Linear Fit

$$\text{Total_Crossings} = -5264.68 + 370.10118 \cdot \text{Avg_Temp}$$

Summary of Fit

RSquare	0.431939
RSquare Adj	0.429208
Root Mean Square Error	4207.555
Mean of Response	14534.5
Observations (or Sum Wgts)	210

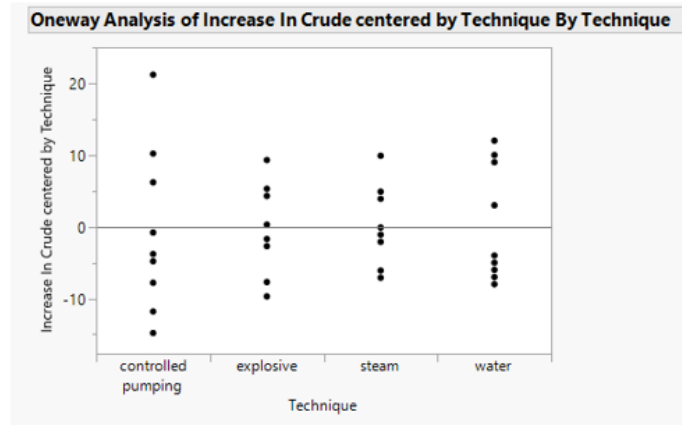
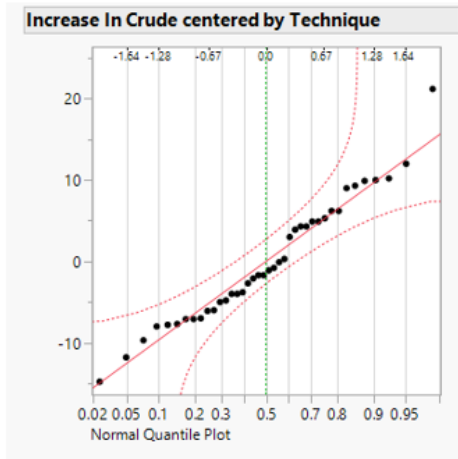
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	2799948519	2.7999e+9	158.1578
Error	208	3682331408	17703516	Prob > F
C. Total	209	6482279927		<.0001*

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-5264.68	1600.902	-3.29	0.0012*
Avg_Temp	370.10118	29.42898	12.58	<.0001*

6. (25 points) A petroleum engineer for a large oil company is evaluating recovery techniques for oil wells. Four techniques are being considered: explosive fracture, water injection, steam injection, and controlled pumping. For each technique the daily increase (barrels) in pumped crude obtained is measured. A one-way ANOVA is run, and the relevant outputs are shown below.



Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Technique	3	371.0000	123.667	1.9781	0.1346
Error	36	2250.6000	62.517		
C. Total	39	2621.6000			

- (a) (3 points) Identify the response for this experiment in the space provided.

Increase in pumped crude

- (b) (3 points) Identify the factor for this experiment in the space provided.

Recovery technique

- (c) (4 points) List all the levels of the factor for this experiment in the space provided.

controlled pumping, explosive, steam, water

(d) (3 points) Select the correct $\mathbf{H}_0$: for the overall F-test.

- ☒ $\mathbf{H}_0$: mean increase in pumped crude is the same for all four techniques
- ☐ $\mathbf{H}_0$: all values for increase in pumped crude are the same for all four techniques
- ☐ $\mathbf{H}_0$: mean increase in pumped crude is the same for at least three of the four techniques
- ☐ $\mathbf{H}_0$: mean increase in pumped crude is the same for at least one of the four techniques

(e) (3 points) Select the correct $\mathbf{H}_A$: for the overall F-test.

- ☐ $\mathbf{H}_A$: mean increase in pumped crude are different for all four techniques
- ☐ $\mathbf{H}_A$: all values for increase in pumped crude are different for all four techniques
- ☐ $\mathbf{H}_A$: mean increase in pumped crude is different for at least three of the four techniques
- ☒ $\mathbf{H}_A$: mean increase in pumped crude is the different for at least one of the four techniques

(f) (4 points) Select the correct conclusion to this test using $\alpha = 0.05$.

- ☐ Reject H_0 . There is enough evidence to conclude the means of all techniques are different
- ☐ Fail to reject H_0 . There is enough evidence to conclude the means of all techniques are the same
- ☐ Reject H_0 . There is enough evidence to conclude the mean of at least one techniques are different
- ☐ Fail to reject H_0 . There is not enough evidence to conclude the means of all techniques are different
- ☒ Fail to reject H_0 . There is not enough evidence to conclude the mean of at least one technique is different

(g) (5 points) Is the normality assumption satisfied?

- ☒ Residuals look good. They follow a straight line and fall within confidence bands
- ☐ Residuals do not look good. They do not follow a straight line and several points fall outside confidence bands
- ☐ Residuals look good as they are evenly spaced out within each technique level